FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – III
PAPER –1
TEST DATE: 25-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A, B, C

Sol. Rays passing through optical centres passes unreflected. Two images are formed at I_1 and I_2

For a convex lens,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

u = –0.4m f = 0.2m

$$= \frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{v}+\frac{1}{0.4}=\frac{1}{0.2}$$

$$\Rightarrow \frac{1}{v}-\frac{1}{0.2}=\frac{1}{0.4}$$

$$\Rightarrow \frac{1}{v}=\frac{1}{0.4}$$

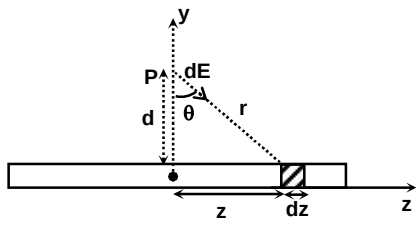
(A) v = 0.4 meter.

(B) Number of images formed = 2

(C) Separation between images is $2\left(3+3\times\frac{0.4}{0.4}\right)$ = 12 mm

2. A, B, C, D

Sol. Gravitational field due to a long wire, $dE = \frac{2G\lambda}{r} = \frac{2G\sigma dz}{r}$



$$E_P = \int dE.\cos\theta = \int \frac{2G\sigma dz}{r}\frac{d}{r} = 2G\sigma d\int_{-\frac{a}{2}}^{+\frac{a}{2}}\left(\frac{dz}{d^2+z^2}\right)$$

$$E_P = 4G\sigma\tan^{-1}\left(\frac{a}{2d}\right)$$

3. A, B, C, D

Sol. $Q = \Delta K$

and use basic concept

4. A, B, C

Sol. $\lambda = 1m$

$\lambda/4 = x$

So, $x = 25$ cm

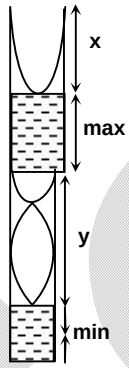
Max $= 125 - 25$

Max $= 100$ cm

$y = 3\lambda/4 = 75$ cm

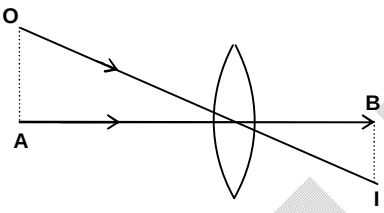
MIN $= 125 - 75 = 50$ cm

$\lambda/2$ = distance between nodes = 50 cm

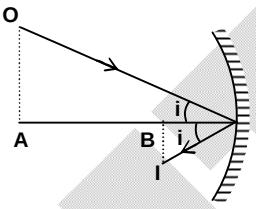


5. A, C, D

Sol. (A)



(C)



(D) Image is inverted $\Rightarrow$ It should be real

6. A, B, C

Sol. Due to a convex mirror of focal length 2.5 cm and due to a concave mirror having its pole at (2cm, 0) real virtual pair

7. D

Sol. Use basic formulae for the given physical quantities to derive the dimensions.

8. D

Sol. $F = +20\,cm$

$u = -30\,cm$

$$v = \frac{uF}{u+F} = \frac{(-30)(20)}{-30+20} = \frac{+600}{-10} = +60\,cm$$

$$m = \frac{v}{u} = \frac{60}{-30} = -2$$

Real, inverted and magnified

$F = -40\,cm$

$u = -30\,cm$

$$v = \frac{(-30)(-40)}{-30+40} = 120\,cm$$

$$m = -\left[\frac{v}{u}\right] = -\left[\frac{120}{-30}\right] = +4$$

virtual, erect and magnified

(R) $\frac{u_2}{v} - \frac{u_1}{u} = \frac{u_2 - u_1}{R}$

$$\frac{1}{v} - \frac{3/2}{-20} = \frac{1-3/2}{-10}$$

$$\frac{1}{v} + \frac{3}{40} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{3}{40} = \frac{2-3}{40} = -\frac{1}{40} \Rightarrow v = -40$$

$$m = +\frac{u_i}{u_r}\frac{v}{u} = \frac{+3/2}{1} \times \left[\frac{-40}{-20}\right] = +3$$

Virtual, erect, magnified

(S) $v = \frac{uF}{u+F} = \frac{(-40)(-40)}{-40-40} = \frac{1600}{-80} = -20$

$$m = \frac{v}{u} = \frac{-20}{-40} = +\frac{1}{2}$$

Virtual, erect, diminished.

9. C

Sol. Use momentum conservation during collision. In case of a plane mirror image and object will be equidistant from the mirror

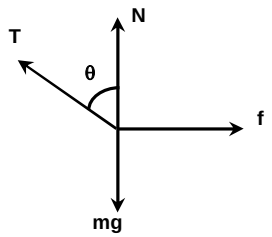
10. D

Sol. $Q = (\Sigma B \cdot E)_{Product} - (\Sigma B \cdot E)_{Reactant}$ **Q**

Section – B

11. 0.24

Sol. Thrust of the ejected water on the tank $T = \rho a u^2$ where u is the speed of the ejected water from tank



$$u = \sqrt{\frac{2gh}{1-(a/A)^2}}$$

Now for the tank to slide, $T\sin\theta \geq \mu(mg - T\cos\theta)$

$$\Rightarrow T \geq \frac{\mu mg}{\sin\theta + \mu\cos\theta}$$

$$\Rightarrow \rho a u^2 \geq \frac{\mu A h \rho g}{\sin\theta + \mu\cos\theta} \qquad \Rightarrow \frac{2(a/A)}{1-(a/A)^2} \geq \frac{\mu}{\sin\theta + \mu\cos\theta}$$

$\Rightarrow \frac{2x}{1-x^2} \geq m$ where $x = \frac{a}{A}$ and $m = \frac{\mu}{\sin\theta + \mu\cos\theta}$

$\Rightarrow x \geq \frac{\sqrt{1+m^2}-1}{m}$ for $\theta = 37°$ and $\mu = 0.5$

$m = 0.5$ and $x \geq \frac{\sqrt{1+0.25}-1}{0.5}$

Hence $x_{min} = 0.236$

12. 0.12

Sol. $P_C = P_D + \rho_2 g h_2 = P_0 + \rho_2 g h_2$

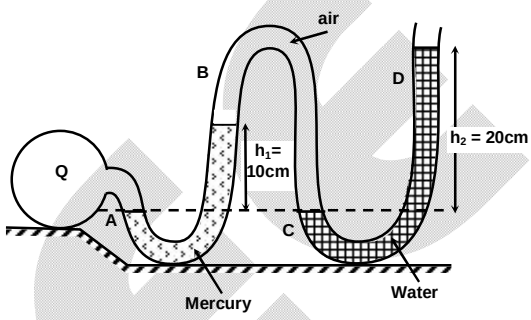
$P_A = P_B + \rho_1 g h_1$

and $P_B = P_C$ (neglecting the density of air as compared to water or mercury)

$\therefore P_A = P_0 + \rho_2 g h_2 + \rho_1 g h_1$

$= 10^5 + 10(13.8 \times 0.1 + 1 \times 0.2) \times 10^3$

$= 10^5 + 10^4 (1.38 + 0.2) = 1.16 \times 10^5 \text{ N/m}^2$



13. 6.00

Sol. Area of element = wdy

$dF = \rho wgydy$

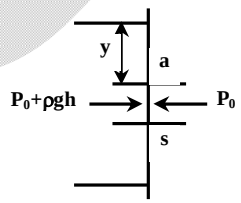
Net force = $\frac{\rho wgD^2}{2}$ . . .(1)

$d\tau = \rho wgydy(D - y)$

$\tau = \frac{\rho wgD^3}{6}$ . . .(2)

Putting the value of F and τ in equation (1) and (2)

D = 10 m and W = 6 m



14. 3.75

Sol. Let V be the total volume of the piece of wood and V′ be its volume inside the liquid. Since the wood piece is in equilibrium

Weight of the wood piece = buoyant force

$\Rightarrow V\rho g = V' \rho_0 g$

$\Rightarrow AL \rho g = AL' \rho_0 g$

- Now if the wood piece is pressed by distance x below the fluid.
  Buoyant force = $A(L'+x) \rho_0 g$
  Net force = $A(L'+x)\rho_0 g - AL \rho g$
  $= AL'\rho_0 g + Ax \rho_0 g - A L \rho g$
  $= A x \rho_0 g$ in the vertically upward direction (Since $A L' \rho_0 g = AL \rho g$)
- Therefore the pressing force = $Ax \rho_0 g$ in the vertically downward direction.
  The work done by the pressing force for a small displacement dx
  $dW = Fdx = Ax \rho_0 g dx$
  Total work done by the pressing force
- $W = \int dW = A \rho_0 g \int_0^{L/2} x\, dx$

$= \frac{A \rho_0 g L^2}{8} = \frac{10^{-2} \times 1200 \times 10 \times (0.5)^2}{8} = 3.75 \text{ J}$

15. 13.33

Sol. $R = V/I = 12/2 = 6\ \Omega$

Error, $\frac{\Delta R}{R}\times 100 = \left(\frac{\Delta V}{V}+\frac{\Delta I}{I}\right)\times 100$

$= \left(\frac{0.4}{12}+\frac{0.2}{2}\right)\times 100$

$= 13.33\%$

$\therefore$ Resistance = $(6\pm 13.33\%)\Omega$

16. 1.53

Sol. Pitch of the Screw gauge = 5/10 mm = 0.5 mm

Least count = $\frac{\text{Pitch}}{100} = \frac{0.5}{100} = 0.005\text{mm}$

Pitch scale reading = Pitch$\times$N = $0.5\times 3 = 1.5$ mm

Difference of the cap reading = F~I = 6

Circular scale reading = Difference $\times$ L.C = 0.03 mm

Diameter = P.S.R. + C.S.R. = (1.5+0.03)mm = 1.53 mm

17. 0.20

Sol. The focal length of the lens is given by

$$\frac{1}{f} = \frac{1}{|v|}+\frac{1}{|u|}$$

substituting the values, we get,

f = 19.9 cm, 20.1 cm, 19.9 cm, 20.2

The average focal length $\cong$ 20.0 cm

The error in a typical measurement:

$$\frac{\delta f}{f^2} = \frac{\delta u}{u^2}+\frac{\delta v}{v^2}$$

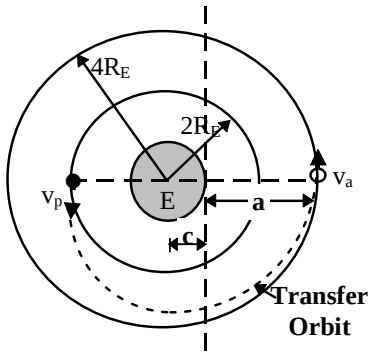
$$\frac{\delta f}{f} = \left(\frac{0.4}{40^2}+\frac{0.4}{(40.5)^2}\right)\times 20 \cong 0.01$$

$\delta f = 0.01 \times 20 = 0.2$ cm

18. 8.00

Sol. Initial orbital speed $v_1 = \sqrt{\frac{GM}{2R}}$ …(i)

Final orbital speed $v_2 = \sqrt{\frac{GM}{4R}}$ …(ii)



For elliptical orbit

From COAM

$v_1' \times 2R = v_2' \times 4R$

$\Rightarrow\ v_2' = \frac{v'}{2}$ …(iii)

From COE $\frac{1}{2}mv_1'^2 - \frac{GMm}{2R} = \frac{1}{2}mv_2'^2 - \frac{GMm}{4R}$

$\Rightarrow\ v_1' = \sqrt{\frac{2GM}{3R}}$

So, energy required = $\frac{1}{2}m\left[\frac{2GM}{3R}-\frac{GM}{2R}\right]+\frac{1}{2}m\left[\frac{GM}{4R}-\frac{1}{4}\frac{2GM}{3R}\right] = \frac{GMm}{8R}$

Chemistry

PART – II

Section – A

19. A, C

Sol. Suppose meq. of NaOH and Na_2CO_3 in the mixture are x and y respectively.

When phenolphthalein is used as indicator

$$x + \frac{y}{2} = 40 \times 0.1 = 4 \quad \ldots \text{(i)}$$

When methyl orange is used as indicator

$$x + y = 60 \times 0.1 = 6 \quad \ldots \text{(ii)}$$

Solving Eq. (i) and Eq. (ii), we get

$x = 2$ meq. and $y = 4$ meq.

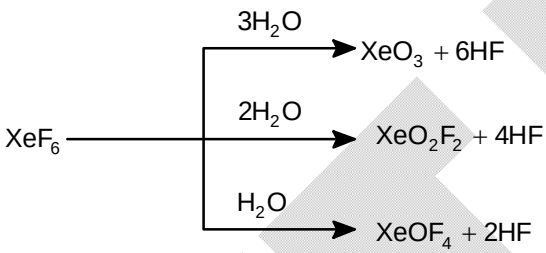
$$\therefore \text{ Weight of NaOH in the mixture} = \left(2 \times 10^{-3} \times 40\right) \times 5 = 0.4 \text{ g}.$$

$$\text{Weight of } Na_2CO_3 \text{ in mixture} = \left(4 \times 10^{-3} \times 53\right) \times 5 = 1.06 \text{ g}$$

$$\text{Weight \% of NaOH in the mixture} = \frac{0.4}{0.4 + 1.06} \times 100 = 27.4\%$$

20. A, B, D

Sol.

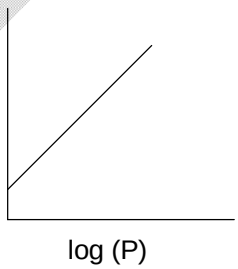


21. B, C, D

Sol. Physisorption is multilayer

log (x/m) vs. log (P)

Slope is +ve.



Physisorption favour at low temperature and high pressure.

$$\left|\Delta H_{\text{physisorption}}\right| < \left|\Delta H_{\text{chemisorption}}\right|$$

22. A, B, D

Sol. Mond's process is used for the purification of Ni.

23. B, C

Sol. $$M_{mix} = \frac{1 \times 16 + 2 \times 4}{1 + 2} = 8$$

$$\frac{r_x}{r_{mix}} = \sqrt{\frac{M_{mix}}{M_x}}$$

$$\Rightarrow 2 = \sqrt{\frac{8}{M_x}} \qquad \therefore M_x = 2$$

$\therefore$ Gas is H_2 gas.

$$\frac{r_{H_2}}{r_{He}} = \sqrt{\frac{4}{2}} = \sqrt{2} \quad \therefore r_x = \sqrt{2}\, r_{He}$$

$\therefore$ (B) and (C) are correct statement.

24. A, C, D

Sol. From given data

$$X_A = \frac{1}{3} \qquad X_B = \frac{2}{3}$$

Where, X_A and X_B are mole fraction of A and B in liquid phase.

If the Y_A and Y_B are mole fraction of And B in vapour phase.

$$\frac{Y_A}{Y_B} = \frac{P_A^o X_A / P_{total}}{P_B^o X_B / P_{total}} = \frac{P_A^o X_A}{P_B^o X_B} = \frac{100 \times \frac{1}{3}}{80 \times \frac{2}{3}} = \frac{5}{8}$$

$$\frac{Y_A}{Y_B} = \frac{n_A}{n_B} = \frac{W_A / M_A}{W_B / M_B} = \frac{W_A}{W_B} \left(\because \text{ A and B are isomer}\right)$$

$\therefore$ Mass ratio of A and B $= \frac{5}{8}$.

$$P_{Total} = P_A^o X_A + P_B^o X_B = 100 \times \frac{1}{3} + 80 \times \frac{2}{3}$$

$$= \frac{260}{3} \text{torr}$$

25. D

Sol. Ziglar Natta $\rightarrow TiCl_4 + Al\left(C_2H_5\right)_3$

Ostwald process $\rightarrow Pt$

Contact process $\rightarrow V_2O_5$

Haber process $\rightarrow Fe$

26. B

Sol. Conceptual

27. B

Sol. Application of Boyle's Law.

28. D

Sol. $\left[Cr\left(H_2O\right)_6\right]^{3+} \rightarrow d^2sp^3 \rightarrow Cr^{+3} \rightarrow d^3 \rightarrow t_{2g}^3 e_g^o$

$\left[Co\left(CN\right)_6\right]^{3-} \rightarrow d^2sp^3 \rightarrow Co^{+3} \rightarrow d^6 \rightarrow t_{2g}^6 e_g^o$

$\left[CuF_6\right]^{4-} \rightarrow sp^3d^2 \rightarrow Cu^{2+} \rightarrow d^9 \rightarrow t_{2g}^6 e_g^3$

$\left[MnF_6\right]^{3-} \rightarrow sp^3d^2 \rightarrow Mn^{+3} \rightarrow d^4 \rightarrow t_{2g}^3 e_g^1$

Section – B

29. 8.00

Sol. $\left[Ag^+\right]_{Anode} = \frac{K_{SP}}{\left[Br^-\right]} = \frac{K_{SP}}{1\times 0.75}M$

$\left[Ag^+\right]_{Cathode} = 0.1\times 0.8 = 0.08\ M$

$E_{cell} = 0.0591\log\frac{\left[Ag^+\right]_{Cathode}}{\left[Ag^+\right]_{Anode}}$

$0.591 = 0.0591\log\left[\frac{0.08}{K_{SP}/0.75}\right]$

$10 = \log\frac{0.75\times 0.08}{K_{SP}}$

$\therefore 10^{+10} = \frac{0.75\times 0.08}{K_{SP}}$

$\therefore K_{SP} = \frac{0.08\times 0.75}{10^{10}} = 6\times 10^{-12}$

Solubility $= \frac{K_{SP}}{\left[Br^-\right]} = \frac{6\times 10^{-12}}{0.75}$ mol / lit.

$x\times 10^{-12} = 8\times 10^{-12}$

$\therefore x = 8$

30. 15.58

Sol. $V_A = \frac{RT}{P_A} = \frac{0.082\times 500}{2} = 20.5$ lit

$A \to B$ is isothermal path

$\therefore V_B = \frac{P_A V_A}{P_B} = \frac{21.5\times 2}{4} = 10.25$ litre

$B \to C$ is adiabatic path

$\frac{P_B}{P_C} = \left(\frac{V_C}{V_B}\right)^{\gamma}$ $\quad \because C_P = \frac{5}{2}R$

$\frac{4}{2} = \left(\frac{V_C}{V_B}\right)^{\frac{5}{3}}$ $\quad \therefore C_V = \frac{3}{2}R$

$\log 2 = \frac{5}{3}\log\frac{V_C}{V_B}$ $\quad \gamma = \frac{5}{3}$

$\Rightarrow \log\frac{V_C}{V_B} = \log 2^{3/5}$

$\frac{V_C}{V_B} = 2^{3/5} = 1.52$

$\therefore V_C = 1.52\times 10.25$

= 15.58 litre

31. 729.00

Sol. $M_{AgX} = \dfrac{\rho \times N_A \times a^3}{4} = \dfrac{\dfrac{40}{6.022} \times 6.022 \times 10^{23} \times \left(6 \times 10^{-8}\right)^3}{4}$

$= 216 \text{ g / mole}$

$\therefore M_X = 216 - 108 = 108 \text{ g / mole}$

If all atoms along one body diagonal have been removed, the formula of unit cell

$= Ag_{4-1}X_{4-\frac{1}{4}} = Ag_3X_{\frac{15}{4}}$

$\therefore$ Mass of 1 mole of these unit cell $= 108 \times 3 + 108 \times \dfrac{15}{4} = 729.00$

32. 4.35

Sol. Number of equivalent of HCl used = Number equivalent of $CaCl_2$ obtained = Number of equivalent of plaster of paris can form $= \dfrac{600 \times 0.1}{1000} = 0.06$

Equivalent weight of plaster of paris $\left(CaSO_4.\dfrac{1}{2}H_2O\right) = \dfrac{\text{Mol.wt.}}{2} = \dfrac{145}{2}$

$\therefore$ Maximum weight of $CaSO_4.\dfrac{1}{2}H_2O$ that can be obtained $= 0.06 \times \dfrac{145}{2} = 4.35 \text{ g}$.

33. 6.25

Sol. $\kappa = \dfrac{1}{R} \times \dfrac{\ell}{A} = \dfrac{1}{50} \times \dfrac{1}{1} = \dfrac{1}{50} \text{ S cm}^{-1}$

$\Lambda_m = \dfrac{\dfrac{1}{50} \times 1000}{0.2} = 100 \text{ S cm}^2 \text{ mol}^{-1}$

$\alpha = \dfrac{\Lambda_m}{\Lambda_m^o} = \dfrac{100}{\lambda^o_{H^+} + \lambda^o_{CH_3COO^-}} = \dfrac{100}{50 + 350} = \dfrac{1}{4}$

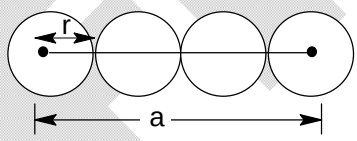
Van't Hoff factor $i = 1 + \alpha = 1.25$

$\pi = i.C.R.T.$

$\pi = 1.25 \times 0.2 \times 300 \times \dfrac{1}{12} = 6.25 \text{ atm}$.

34. 78.50

Sol. $a = 6r$



Effective number of atom in one unit cell $= 4 \times \dfrac{1}{4} + 8 \times \dfrac{1}{2} + 4 = 9$

$\therefore \%P.E. = \dfrac{9 \times \pi r^2}{a^2} \times 100$

$= \dfrac{9 \times 3.14 \times r^2}{(6r)^2} \times 100$

$= \frac{9 \times 314r^2}{36r^2} = \frac{314}{4}$

= 78.5%

35. 0.70

Sol. Amount of thermal energy required $= w \times C_S \times \Delta T$

Amount of thermal energy in beaker A $= 200 \times C_S \times (90 - 40) = 10000\ C_S$

Amount of thermal energy in beaker B $= 100 \times C_S \times (90 - 20) = 7000\ C_S$

Amount of thermal energy in beaker C $= 150 \times C_S \times (90 - 50) = 6000\ C_S$

Amount of thermal energy in beaker D $= 300 \times C_S \times (90 - 30) = 18000\ C_S$

$\therefore w = 6000\ C_S,\ x = 7000\ C_S,\ y = 10000\ C_S,\ z = 18000\ C_S$

$\therefore \frac{x}{y} = \frac{7000\ C_S}{10000\ C_S} = 0.7$

36. 9.00

Sol. Weight of oxalic acid in 50 ml solution before adsorption $= \frac{45}{500} \times 50 = 4.5$ gm

Weight of oxalic acid in 50 ml solution after adsorption $= 0.6 \times \frac{50}{1000} \times 90 = 2.7$ gm

$\therefore$ Weight of oxalic acid adsorbed by 0.2 gm of charcoal = 4.5 – 2.7 = 1.8 g

$\therefore$ Weight adsorbed per gram $= 1.8 \times \frac{1}{0.2} = 9$ gm.

Mathematics

PART – III

Section – A

37. A, B, C

Sol. $P_4 = \frac{{}^4C_2}{4^4}$, $P_5 = 0$, $P_6 = \frac{2(60-12)}{4^6} = \frac{96}{4^6}$

38. A, D

Sol. $|\vec{P}_1| = \sqrt{3}$, $|\vec{P}_2| = 1$ and $\vec{P}_1 \cdot \vec{P}_2 = 0 \Rightarrow |\vec{P}_3 \cdot \vec{P}_1 \times \vec{P}_2| = |\vec{P}_3||\vec{P}_1||\vec{P}_2|$

39. C, D

Sol. $\frac{m}{n} = \frac{101}{51}$, $N = \frac{(m-1)(n-1)}{2}$

40. A, C, D

Sol. $\frac{(z^3 - z^2)}{(z - z^2)} \cdot \frac{(z - z^4)}{(z^3 - z^4)} \in R \Rightarrow (z - \bar{z})(z\bar{z} - 1) = 0$ for given order z, z^2, z^3, z^4

$\Rightarrow \arg z \in \left(0, \frac{2\pi}{3}\right) \cup \left(\frac{4\pi}{3}, 2\pi\right)$

41. A, C, D

Sol. $\det(M_1) = \det(M_2) = 0 \Rightarrow \det(aM_1 + bM_2) = 0 \ \forall \ a, b \in R$

42. A, B

Sol. z_1, z_2, z_3 are the roots of the equation $z^3 - 2\lambda z^2 + 2\lambda^2 z - \lambda^3 = 0$

43. B

Sol. $n(A) > n(B) \rightarrow$ Total number of cases 211

$n(A) = n(B) \rightarrow$ Total number of 32 cases

44. D

Sol. $(a + b + c)((a - b)^2 + (b - c)^2 + (c - a)^2) = 2$

If $a, b, c \in Z \Rightarrow (1, 0, 0), (0, 1, 0), (0, 0, 1)$ are solutions

If $t \in \theta \Rightarrow a = c$, $2a + b = \frac{1}{t^2}$, $a - b = t \Rightarrow$ infinite possibility

45. D

Sol. $|((z - 2)^8 - 3^8| = |(z - z_1)(z - z_2)(z - z_3) ..... (z - z_8)|$

46. C

Sol. (I) There are 4 distinct path

(II) There are $\frac{1}{3}\left({}^{10}C_5\right)$ distinct path

(III) Count rectangle of all possible dimension = 73

(IV) There are 218 distinct path

Section – B

47. 95.00

Sol. Let θ_i be the event that i is the upper left corner of required square

$$\Rightarrow P(\theta_i \cup \theta_2 \cup \theta_3 \cup \theta_4) = \frac{95}{512} \Rightarrow P(\theta_1^c \cap \theta_2^c \cap \theta_3^c \cap \theta_4^c) = \frac{417}{512}$$

48. 81.00

Sol. Suppose there are T_n ways to cover n steps

$T_n = T_{n-1} + T_{n-2} + T_{n-3}$, $T_1 = 1$, $T_2 = 2$

49. 220.00

Sol. Let $p = 9 - a$, $q = 9 - a_3 \Rightarrow p + a_2 + q + a_4 = 9 \Rightarrow {}^{12}C_3 = 220$

50. 4.00

Sol. $|A^2 + |A|A + Tr(A)\cdot I_2| = 4(2(|A| - 1)^2 + 1)$

51. 23.00

Sol. $P(x) = x^3 + 2x^2 + 3x + 3$

52. 3.00

Sol. $\left(\frac{\vec{p}+\vec{q}+\vec{r}}{3}\right)\cdot\left(\frac{\vec{p}+\vec{r}}{3}-\vec{q}\right) = 0 \Rightarrow |\vec{p}|^2 + |\vec{q}|^2 = 3|\vec{r}|^2$

53. 2.78

Sol. Let $\alpha = u$, $\beta = u + v$, $\gamma = u + 2v$, $\delta = u + 3v \Rightarrow p - q = -\frac{1}{9}(u+1)^2(u^2 + 2u - 9)$

54. 834.00

Sol. Let $im = n \rightarrow 9 \times 34$ cases

$im^5 = n^2 \rightarrow 8 \times 33$ cases

$im^9 = n^3 \rightarrow 8 \times 33$ cases